

Portfolio Project Brief

Cloud Computing and Scalable Architectures Advanced Software Engineering - Module 6

How to use this brief:

Read this document in full and bring any questions you have on the assessment to either a workshop or the next 1:1 with your instructor. **Advance in your project journey by completing milestones**, which are progress checks on tasks that build to your overall output. Milestone submission and reviews will be part of your workshop and independent work. **Self-assess your progress** against the assessment criteria table while completing your project.

Module Title	Cloud Computing and Scalable Architectures		
Format	Written Report	Deadline	On My Multiverse
Length	4000 words	Feedback Due	Within 20 working days of submission date
Project Title	Create a Cloud-Based Tool to Enhance Architecture		
Project Learning Outcomes	1. Explain fundamental computer networking concepts in relation to cloud architecture, structure, components, and quality of service (K16). 2. Interpret and implement a design compliant with functional, non-functional and security requirements for cloud-based solutions (K24). 3. Respond to changing priorities and problems arising within software engineering projects by making revised recommendations and adapting plans as necessary (S20). 4. Describe how tools that support teamwork can be used effectively in cloud platform development (K28).		
Project Milestones	Milestone 1: Conduct a comprehensive evaluation of your organisation's use of cloud computing technologies, principles, and architectures. Based on your evaluation, determine your organisation's functional, non-functional and security requirements (completed by week 3). Milestone 2: Design and build a cloud-based tool that supports your organisation's functional, non-functional and security requirements related to cloud computing. Consider factors such as performance, flexibility, resource optimisation, and scalability (completed by week 7).		

	Milestone 3: Assess the impact of the tool within your organisation and provide recommendations to further enhance architecture resiliency and scalability with cloud-based solutions (completed by week 10).
Feedback	Written feedback and a grade will be given within 20 working days after the deadline. Opportunity for further discussion will be available during individual coaching sessions.

Final Project Output

Your final project output will be a written report that documents the design, development, and implementation considerations for a cloud-based tool within your organisation. The report should demonstrate your understanding of cloud computing principles, your ability to design compliant architectures, and your capacity to respond to changing priorities within software engineering projects.

Suggested Report Structure

The structure below provides suggested guidance on content coverage and approximate word counts. You may adapt the section titles and organisation to best suit your project, provided all Learning Outcomes are evidenced in your submission.

Section	Content Covered
Introduction & Overview (800 words)	- Explain fundamental computer networking concepts in relation to cloud computing (e.g., structure, cloud architecture, components, quality of service) (K16). - Provide an overview of the cloud-based tool you will design and build, including its function and alignment to the software development lifecycle (K16).
Design Cloud-Based Tool (1200 words)	- Interpret your organisation's functional, non-functional, and security requirements (K24). - Design a cloud-based architectural solution that addresses these requirements (K24). - Explain how your design considers performance, flexibility, resource optimisation, and scalability (K24). - Describe how your design could adapt to changing organisational priorities (S20).
Develop Cloud-Based Tool (1200 words)	- Implement your cloud-based tool design, documenting techniques and methods used (K24). - Describe how tools that support teamwork (e.g., configuration management, version control, release management) were or could be used in the development process (K28). - Test your tool's performance, flexibility, resource optimisation, and scalability (K24).

Conclusion & Recommendations (800 words)	• Provide recommendations for implementing your cloud-based tool. • Summarise your project's goals, methodology, and anticipated ROI.
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Project Milestones

Project milestones are completion points for the overall output of your module project. Please note that the project is intended to be a piece of applied work that develops over the course of the module, rather than a summative assessment to be completed in a short, intense period of time at the end of module learning.

NOTE: For step-by-step guidance on milestone completion, follow the Milestone activities for the module on your MyMultiverse account.

How should I submit this Portfolio Project?

Your submission should be prepared as a single PDF file. Please make sure your work adheres to the MV referencing and style guide (see MyMultiverse). All work is checked for plagiarism using Turnitin.

Ensure that you meet the MV guidelines for academic integrity by considering the following:

- **Referencing:** When discussing someone else's work, clearly acknowledge the work by using in-text citations and a reference list adhering to the Harvard referencing style.
- **Word Count:** Any text exceeding the maximum word limit will not be marked. Text that adds to the word count includes all text except the reference list or appendices. All text in diagrams and in-text citations count towards the overall word limit.
- **Submission time:** All submissions must be made by mid-day on the specified deadline.

How will my assessment be marked?

Carefully review the assessment criteria as you complete milestones and outputs of your project.

	F	E	D	C	B	A
LO1: K16	Does not explain computer networking concepts. Shows no engagement with cloud architecture or quality of service.	Identifies networking concepts but fails to describe their relationship to the cloud-based tool's structure.	Describes fundamental networking concepts (e.g. TCP/IP, subnets). Explains how these components form the structure of the proposed cloud architecture.	Explains how networking concepts (e.g. Load Balancing, Quality of Service) inform specific design decisions for the tool.	Analyses the interaction between networking components within the organisational environment. Justifies the architecture using context-specific data.	Evaluates alternative architectural patterns. Critically appraises how networking selections (e.g. Latency vs. Throughput) address strategic organisational requirements.
LO2: K24	Does not interpret requirements or implement a design for the cloud-based solution.	Identifies basic requirements but fails to implement a design that addresses security or non-functional needs (e.g. scalability).	Interprets functional, non-functional, and security requirements. Implements a cloud design that addresses these for a specific organisational tool.	Justifies design decisions by explaining how specific cloud components (e.g. IAM, Auto-scaling) address the interpreted requirements. Documents basic performance testing.	Analyses trade-offs between design alternatives (e.g. Serverless vs. Containers). Synthesises testing data to evaluate quality controls and resource optimisation.	Critically evaluates design effectiveness across multiple failure scenarios. Justifies the selection of architectural patterns against legacy system constraints.
LO3: S20	Does not describe how to respond to changing priorities or provide revised recommendations.	Identifies a problem but provides minimal evidence of adapting the project plan to fit the new scenario.	Describes a response to at least one changing priority. Makes a revised recommendation and adapts the project plan accordingly.	Explains the response to two or more changing priorities. Justifies the revised plan with clear technical reasoning for the chosen adaptation.	Analyses different approaches to responding to change. Demonstrates how adapted plans influenced team outcomes or project deliverables.	Critically evaluates the implications of adapted plans on project success. Compares and contrasts adaptation strategies, justifying the most resilient approach and considering future implications.

LO4: K28	Does not describe tools that support teamwork or their use in cloud platform development.	Identifies tools (e.g. Git, Jenkins) but provides an incomplete account of their use in the development process.	Describes how at least two teamwork tools (e.g. Version Control and Release Management) are used in cloud platform development.	Explains how teamwork tools enhance collaboration effectiveness. Demonstrates their use within the specific context of the cloud-based tool development.	Analyses how tools support distributed teams and code integration. Justifies the use of specific configuration management approaches (e.g. Infrastructure as Code).	Evaluates the effectiveness of the teamwork toolchain. Compares alternative tools and justifies selections based on their impact on team productivity and cloud code quality.
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Knowledge, Skills and Behaviours (KSB) descriptions

Code	Description
K16	Fundamental computer networking concepts in relation to digital and technology solutions. For example, structure, cloud architecture, components, quality of service.
K24	How to interpret and implement a design, compliant with functional, non-functional and security requirements including principles and approaches to addressing legacy software development issues from a technical and socio-technical perspective. For example, architecture, languages, operating systems, hardware, and business change.
K28	Approaches to effective teamwork and the range of software development tools supporting effective teamwork. For example, configuration management, version control and release management.
S20	Respond to changing priorities and problems arising within software engineering projects by making revised recommendations, and adapting plans as necessary, to fit the scenario being investigated.